Power System Design and Performance Analysis of an M100 Methanol Range Extended Electric Dump Truck

Xiqing Zhang, Yanbo Dai, Yafeng Ren, Yaojie Wang, and Shiwei Zhao

Abstract Dump trucks typically utilize diesel fuel, resulting in significant emissions of nitrogen oxides and particulate matter in their exhaust gas, which contributes to substantial environmental pollution. To support the strategy for achieving carbon peak and carbon neutrality, this study focuses on the range extended electric dump truck, using methanol as an alternative fuel to diesel. After completing the performance test of the M100 methanol range extender, the operating point of the range extender is determined. Considering the design requirements and performance indicators of the dump truck, the parameter design for both the power system and drive system is completed, and the 4-speed AMT is matched with the drive motor. The acceleration performance and fuel economy of the dump truck are simulated using AVL CRUISE. The results indicate that the acceleration time from 0 to 50 km/h is 7.73 s, and the maximum gradeability is 45.92%, the acceleration performance satisfies the design requirements. Under the CHTC-D cycle, the dump truck methanol consumption is 70.66 L/100 km, the electric energy consumption is 1.05 kWh/km, the equivalent diesel consumption is 29.79 L/100 km, and the driving range is 425 km, the fuel economy also satisfies the design requirements. The research on acceleration performance and fuel economy of the M100 methanol range extended electric dump truck lays a foundation for the future formulation and optimization of the control strategy for the range extender.

Keywords Range extended electric dump truck · Methanol range extender · Acceleration performance · Fuel economy

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1 Introduction

The dump truck is a kind of widely used commercial vehicle. With its compact body design and high cargo capacity, it has played a significant role in transporting urban waste soil and transferring building materials. The dump truck has a large load capacity and typically operates at low speed in low gear. However, this results in a low engine load rate and high fuel consumption rate.

Range Extended Electric Vehicle (REEV) is powered by the drive motor, while the range extender is not directly involved in vehicle drive. It only charges the power battery when its State of Charge (SOC) is low [1], ensuring that the range extender consistently operates in the high efficiency range. Additionally, due to the high torque characteristics of the motor at low speed, it can fully utilize its driving force to satisfy the operating conditions of the dump truck. Manfredi et al. [2] shown that the selection of cell chemistry and the constraints on battery weight can affect battery performance and the optimal energy management strategy of the REEV. Parkar et al. [3] developed a Modified Particle Swarm Optimization (Modified PSO) methodology aimed at minimizing fuel consumption and NOx emissions of the REEV. Through the coordination and energy transfer between the range extender, power battery, and drive motor, the range extended electric dump truck can significantly reduce fuel consumption and achieve better fuel economy compared to traditional fuel-powered dump trucks.

Methanol offers several advantages, including environmental protection, low exhaust pollution, low cost, and high safety. China's energy structure is characterized by an abundance of coal, a scarcity of oil, and limited natural gas resources. High-sulfur coal, which constitutes over 40% of the country's coal reserves, is not suitable for direct use as fuel in industrial applications such as power generation. However, it has been widely utilized in the production of industrial methanol. Currently, coal-derived methanol remains the predominant type of methanol produced [4]. The development of methanol engines can fully leverage China's coal resources and aligns with the country's basic national conditions.

From 1970s, China began researching methanol engines. Currently, it has achieved several technological breakthroughs and a small range of applications in passenger vehicles and commercial vehicles. Efforts are underway to carry methanol engines on construction machinery and large ships. However, there is relatively little research on methanol range extenders. Methanol engines face challenges such as high fuel consumption and low thermal efficiency [5]. Integrating a generator to create a vehicle-mounted methanol range extender can effectively address the existing technical issues associated with methanol engines while also compensating for the limited driving range of electric dump trucks.

Table 1 Basic parameters of the dump truck

Parameters	Numeric value
Drive form	6 × 4
Overall dimension/mm	7400 × 2550 × 3300
Wheelbase/mm	3800 + 1400
Curb weight/kg	14,000
Gross weight/kg	25,000
Frontal area/m^2	8.0
Drag coefficient	0.70
Final drive ratio	7.63
Methanol tank capacity/L	300
Tire size	12.00R20 16PR
Tire rolling radius/mm	545.4

2 Basic Structure of the Dump Truck

2.1 Basic Parameters of the Dump Truck

The dump truck is designed with a gross weight of 25 tons and adopts a 6 × 4 drive form. The basic parameters are shown in Table 1.

2.2 Performance Measures of the Dump Truck

The dump truck typically operates at low velocity in urban and rural areas. Due to its gross weight and operating conditions, the power requirement is significant. To accommodate various operating environments, the performance design measures of the dump truck are shown in Table 2.

Table 2 Performance measures of the dump truck

Measures	Numeric value
Maximum velocity/(km/h)	90
Acceleration time from 0 to 50 km/h/s	8.0
Maximum gradeability/%	45
Electric energy consumption/(kWh/km)	< 1.2
Methanol consumption/(L/100 km)	< 80

3 Design of Power and Drive Systems

3.1 Performance Test of the Range Extender

Based on an L6 natural gas engine, a spark ignition M100 methanol engine has been developed. The main technical parameters of the engine are shown in Table 3.

The M100 methanol engine serves as the power source, paired with a rated power of 150 kW Permanent Magnet Synchronous Motor (PMSM) to create the range extender for the dump truck. The main technical parameters of the range extender are shown in Table 4.

Table 3 Main technical parameters of the M100 methanol engine

Parameters	Numeric value
Type	4-valve/4-stroke/L6
Emission standard	China VI
Ignition mode	Spark plug ignition
Capacity/L	6.87
Cylinder bore/mm	108
Cylinder stroke/mm	125
Compression ratio	11.5:1
Intake form	TurboCharger + InterCooler
Spray form	Inlet multi-point electrojet
Rated power/kW	190
Rated torque/Nm	1000
Rated speed/(r/min)	2100
Technical route	Theoretical air–fuel ratio + EGR + TWC

Table 4 Main technical parameters of the range extender

Parameters	Numeric value
Rated power/kW	150
Rated voltage/V	540 V DC
Rated current/A	270
Rated frequency/Hz	50 / 60
Rated speed/(r/min)	1500 / 1600
Excitation mode	Brushless excitation
Voltage regulation range	$\geq 5\%$
Voltage fluctuating range	$\leq \pm 5\%$
Insulation class	H

The Zero-Carbon Power Comprehensive Test Platform (CJ560) is designed with a maximum test power of 560 kW, enabling a thorough and accurate evaluation of the acceleration performance and fuel economy of the 150 kW M100 methanol range extender. The arrangement of the range extender on the bench as shown in Fig. 1 and the test environment as shown in Fig. 2. The Zcanpro software communicates with the Electronic Control Unit (ECU) and the Motor Control Unit (MCU) via CAN communication to control the speed and torque of the range extender. Test data is measured through various instruments, including the YOKOGAWA WT1804E Power Analyzer and the CAMA Methanol Consumption Meter. The universal characteristics of the M100 methanol engine and range extender are shown in Fig. 3. The main performance parameters of the range extender are summarized in Table 5.

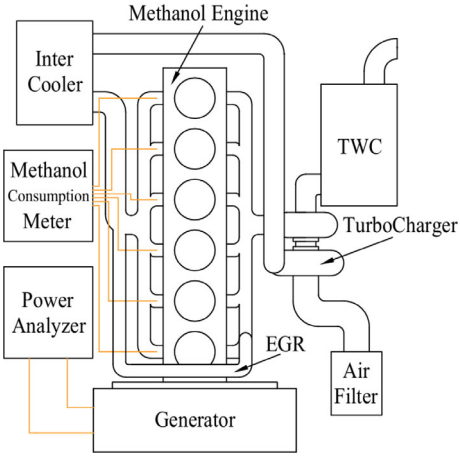


Fig. 1 Arrangement of the range extender on the bench

Fig. 2 Test environment

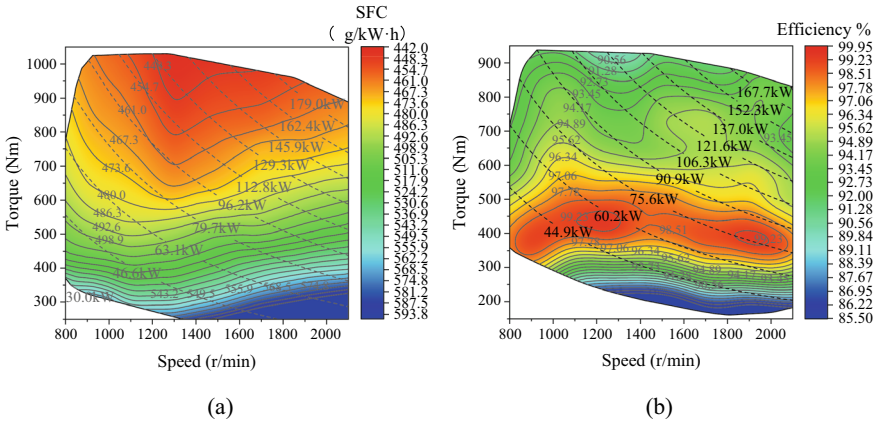


Fig. 3 Test result **a** Universal characteristic of the M100 methanol engine; **b** Universal characteristic of the range extender

Table 5 Main performance parameters of the range extender

Parameters	Numeric value
Maximum power/kW	182.57
Maximum torque/Nm	991.83
Minimum methanol consumption rate/(g/kWh)	436.72
Maximum thermal efficiency/%	35.85
Maximum power generation efficiency/%	99.93

3.2 Design of Drive Motor Parameters

The drive motor of the range extended electric dump truck must provide stable torque output at low speed and stable power output at high speed. The type of drive motor utilized is PMSM. P_{emax}, the peak power of the drive motor should satisfy the following conditions.

(1) P_{e1}, the demand power of the drive motor at maximum velocity.

When the range extended dump truck operates at maximum velocity in a uniform straight line on well-leveled pavement, both grade resistance and acceleration resistance can be neglected. The power of the drive motor should satisfy Eq. (1).

$$P_{e1} = \frac{v_{max}}{3600\eta_t}\left(m_f gf + \frac{C_D A v_{max}^2}{21.15}\right) \tag{1}$$

In the above equation, v_{max} is the maximum velocity 90 km/h; η_t is the transmission efficiency 0.95; m_f is the gross weight of the dump truck 25,000 kg; g is the gravitational acceleration 9.8 m/s^2; f is the tire rolling resistance coefficient 0.02; C_D is the drag coefficient 0.7; *A* is the frontal area 8 m^2. P_{e1} = 185.39 kW is obtained.

(2) P_{e2}, the demand power of the drive motor at maximum gradeability.

When the range extended dump truck operates at full load and uniform velocity on maximum grade road, the acceleration resistance can be neglected. The power of the drive motor should satisfy Eq. (2).

$$P_{e2} = \frac{v_\alpha}{3600\eta_t}\left(m_f g f \cos\alpha + \frac{C_D A v_\alpha^2}{21.15} + m_f g \sin\alpha\right) \tag{2}$$

In the above equation, v_α is the uniform climbing velocity 8 km/h; $\tan\alpha$ is the maximum gradeability 45%. $P_{e2} = 245.44$ kW is obtained.

The peak power of the drive motor should satisfy Eq. (3). P_{emax} is rounded, $P_{emax} = 250$ kW.

$$P_{emax} \geq \max[P_{e1}, P_{e2}] \tag{3}$$

P_{e0}, the rated power of the drive motor should satisfy Eq. (4).

$$P_{e0} = \frac{P_{emax}}{\lambda} \tag{4}$$

In the above equation, λ is the motor overload coefficient 1.6. P_{e0} is rounded, $P_{e0} = 160$ kW.

The relationship between the torque and power of the motor is described by Eq. (5).

$$T_e = \frac{9550 P_e}{n_0} \tag{5}$$

High power PMSM for vehicles is typically a low-speed motor. In the above equation, n_0 is the rated speed of drive motor 1500 r/min; both the peak torque T_{emax} and the rated torque T_{e0} of the drive motor are obtained and rounded, $T_{emax} = 1600$ Nm, $T_{e0} = 1000$ Nm.

3.3 Design of Transmission Parameters

(1) i_{gmax}, the slowest ratio of transmission should satisfy Eq. (6) and Eq. (7).

$$i_0 i_{g\,max} \geq \frac{r}{T_{e\,max}\eta_t}\left(m_f g f \cos\alpha + \frac{C_D A v_\alpha^2}{21.15} + m_f g \sin\alpha\right) \tag{6}$$

$$i_0 i_{gmax} \leq \frac{0.377 r n_{max}}{v_\alpha} \tag{7}$$

In the above equation, i_0 is the final drive ratio 7.63; r is the tire rolling radius 545.4 mm; n_{max} is the maximum speed of the drive motor 3400 r/min. $4.93 \leq i_{gmax} \leq 11.45$ is obtained.

(2) i_{gmin}, the top gear ratio of transmission should satisfy Eq. (8) and Eq. (9). $0.33 \leq i_{gmin} \leq 1.02$ is obtained.

$$i_0 i_{gmin} \geq \frac{r}{T_{emax}\eta_t}\left(m_f g f + \frac{C_D A v_{max}^2}{21.15}\right) \tag{8}$$

$$i_0 i_{gmax} \leq \frac{0.377 r n_{max}}{v_{max}} \tag{9}$$

According to engineering conditions, the transmission is designed to have four forward gears. Select 1 gear ratio $i_{g1} = i_{gmax} = 5.0$, 4 gear ratio $i_{g4} = i_{gmin} = 1.0$. To ensure that the drive motor operates in the high efficiency range, select $i_{g2} = 3.05$, $i_{g3} = 1.66$.

3.4 Design of Power Battery Parameters

The power battery directly supplies energy to the drive motor, which necessitates a high energy conversion rate. To minimize both the size of the power battery and weight of the dump truck, the onboard power battery pack must possess higher specific energy and specific power. The range extended electric dump truck experiences frequent charging and discharging cycles, requiring the power battery to support a greater number of charge and discharge cycles. Consequently, Lithium Iron Phosphate Battery has been chosen as the power battery for the range extended electric dump truck.

The dump truck is equipped with the range extender that starts when the power battery SOC is low. Consequently, when determining the capacity of the power battery, it is not necessary to consider the driving range of the dump truck, it should only satisfy the demand power of the drive motor in Eq. (10).

$$C_b \geq \frac{1000}{kU_m}\left(\frac{P_{emax}}{\eta_e} + P_A\right) \tag{10}$$

In the above equation, k is the maximum discharge rate of the battery 6.5 h^{-1}; U_m is the bus voltage 650 V; η_e is the efficiency of drive motor 0.9; P_A is the accessory power of the dump truck 20 kW. $C_b \geq 70.48$ kWh is obtained. Considering that reducing the frequency of battery charging and discharging can significantly enhance the service life of the power battery, select $C_b = 150$ kWh.

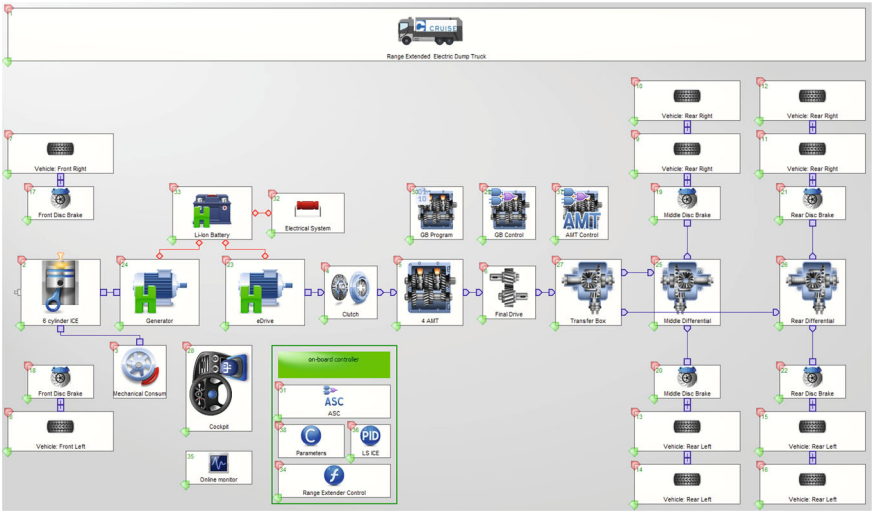


Fig. 4 Simulation model of the dump truck

4 Model Building and Simulation Analysis of the Dump Truck

4.1 Model Building of the Dump Truck

AVL CRUISE can simulate and analyze the acceleration performance and fuel economy of various types of vehicles, including traditional fuel vehicles and new energy vehicles. Utilizing the hybrid electric vehicle module in AVL CRUISE, the model of the range extended electric dump truck, as shown in Fig. 4, has been built. The parameters for the vehicle, engine, generator, power battery, drive motor, transmission, final drive, differential, brake, tire, and other components have been established and interconnected. The blue lines in Fig. 4 represent mechanical connections between components, while the red lines represent electrical connections, with components passing signals via the data bus.

4.2 Running Resistance of the Dump Truck

According to reference [6], the running resistance of the vehicle can be determined through the vehicle sliding test, as well as by Eq. (11).

$$v = \frac{m_f \times 2 \times \Delta v}{3.6 \times \left(A + Bv + Cv^2\right)} \tag{11}$$

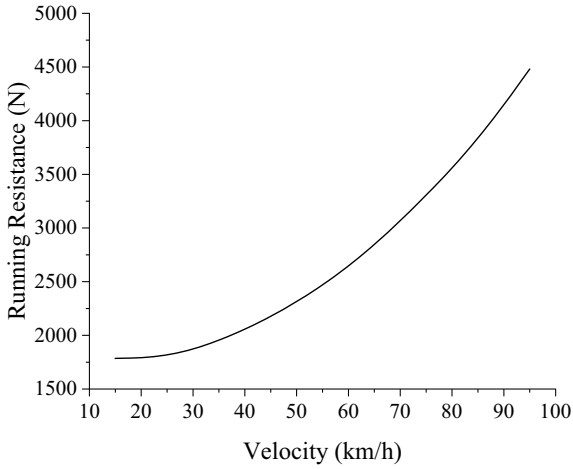


Fig. 5 Running resistance of the dump truck

In the above equation, v is the running velocity of the dump truck, which ranges from 15 km/h to 95 km/h; Δv is the deviation from velocity 5 km/h; A is the constant term of the running resistance 1677.2N; B is the monomial coefficient 13.74 N/(km/h); C is the quadratic coefficient 0.246 N/(km/h)^2. The running resistance of the dump truck is obtained by fitting as shown in Fig. 5.

4.3 Control Strategy of the Range Extender

(1) Operating point select of the range extender.

The rated power of the range extender is 150 kW. To achieve better fuel economy and emission, the operating point of the range extender is determined based on the performance test in Sect. 3.1. At this operating point, the speed is 1450 r/min, the torque is 988 Nm, the thermal efficiency is 35.6%, and the power generation efficiency is 93.84%.

The Function Module of AVL CRUISE is used to control the operating point of the range extender. To enhance the service life of the power battery, the range extender starts to charge the battery when the SOC falls below 30%. Conversely, when the SOC exceeds 80%, the range extender is turned off.

(2) Shift strategy of the transmission.

Among the various types of automatic transmissions, AMT offers several advantages, including high transmission torque, excellent efficiency, remarkable reliability, and low cost. Additionally, the shifting process is managed through electronic control, simplifying the operation for drivers and making it a popular choice for trucks. In AVL CRUISE, the AMT Control, Gearbox Control, and Gearbox Program are utilized to

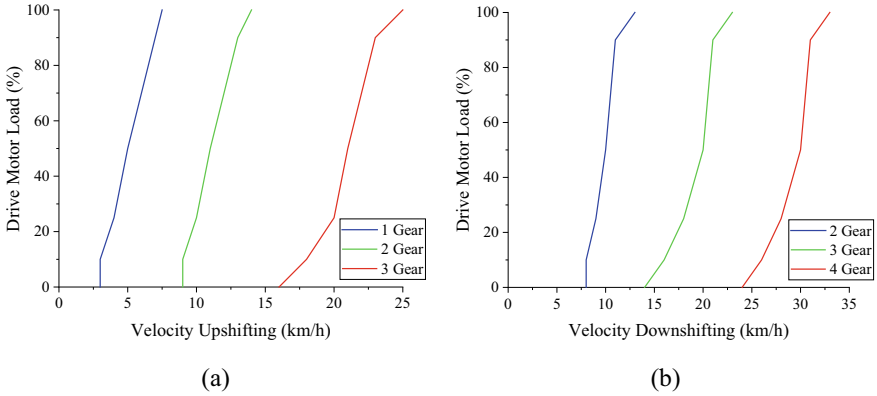


Fig. 6 Shift strategy of the AMT; **a** Upshift strategy; **b** Downshift strategy

control the operation of the 4-speed AMT. The transmission shift strategy is shown in Fig. 6.

4.4 Simulation Analysis

(1) Acceleration performance simulation of the dump truck.

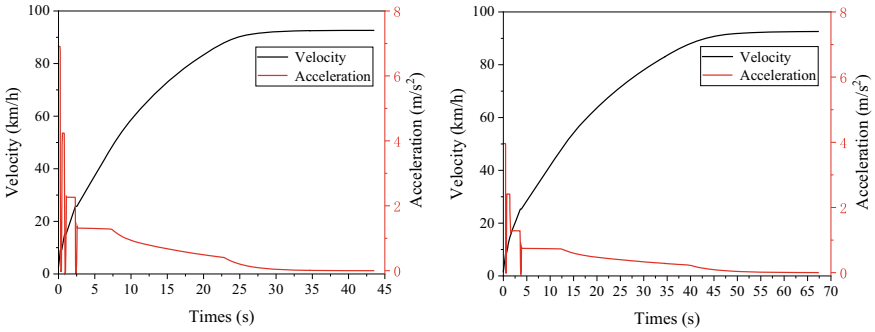
The simulation results for the maximum velocity of the dump truck under no load are shown in Fig. 7a. The maximum velocity is 92.63 km/h and the acceleration time from 0 to 50 km/h is 7.73s. This performance satisfies the requirements both maximum velocity is 90 km/h and the acceleration time from 0 to 50 km/h is 8s. The simulation results for the maximum velocity of the dump truck under full load are shown in Fig. 7b. The maximum velocity is 92.59 km/h, with an acceleration time from 0 to 50 km/h of 13.17s.

The simulation results for the maximum gradeability of the dump truck under full load are shown in Fig. 8. The maximum gradeability in 1 gear is 45.92%, which satisfies the maximum gradeability requirement of 45% for the dump truck.

(2) Fuel economy of the dump truck.

CHTC-D is the latest dump truck cycle in China, featuring a total operating time of 1300 s and an operating range of 8.37 km. Within this cycle, the low-velocity condition lasts for 540 s, while the high-velocity condition lasts for 760 s. The maximum velocity is 71.4 km/h, and the average velocity is 23.19 km/h. The velocity curve for the CHTC-D cycle is shown in Fig. 9.

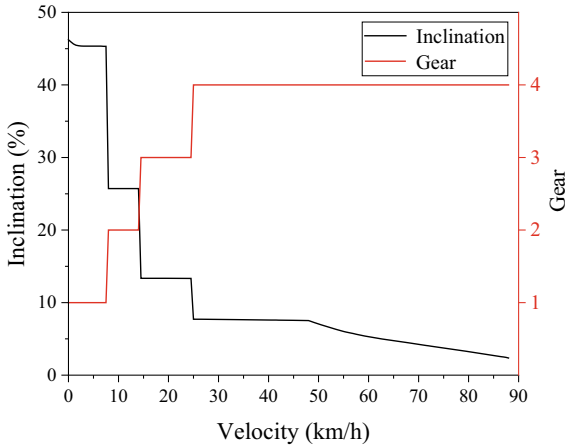
As shown in Fig. 10, a total of 33 CHTC-D cycles are conducted. The dump truck accumulates an operating range of 276.21 km. The initial SOC of the power battery is 80%. A total of three charge and discharge cycles are completed. The range extender



(b)

Fig. 7 Maximum velocity of the dump truck; **a** No load; **b** Full load

Fig. 8 Maximum gradeability of the dump truck



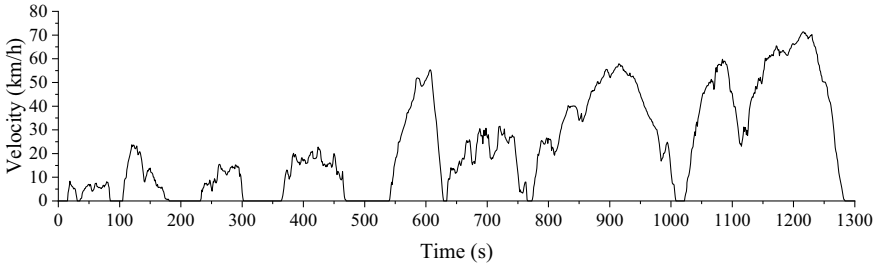


Fig. 9 CHTC-D cycle

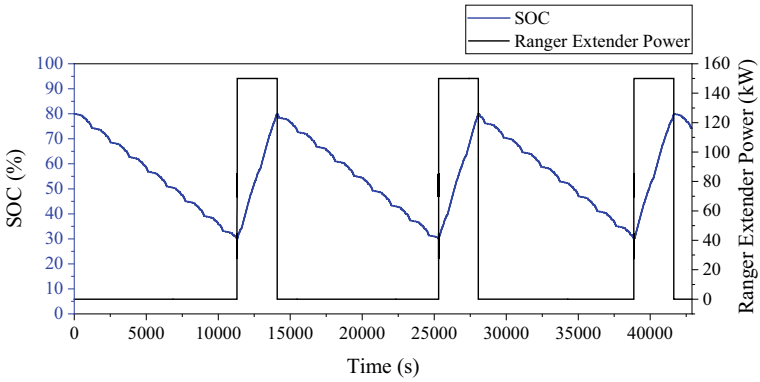


Fig. 10 CHTC-D SOC

is started three times, providing a stable power supply to the battery at 150 kW. At the end of the simulation, the battery SOC is 74.25%.

C-WTVC is developed by China based on the World Transient Vehicle Cycle (WTVC). It is applicable to various types of heavy commercial vehicles. The total operating time is 1800s, with an operating range of 20.52 km. This includes an urban condition of 900 s, a highway condition of 468 s, and a high-velocity condition of 432 s, with a maximum velocity of 87.8 km/h. The velocity curve for the C-WTVC cycle is shown in Fig. 11.

As shown in Fig. 12, a total of 16 C-WTVC cycles are conducted. The dump truck accumulates an operating range of 328.32 km. Similar to the CHTC-D cycle, the initial SOC of the power battery is 80%. A total of three charge and discharge cycles are completed. The range extender is started three times and the power is 150 kW. At the end of the simulation, the battery SOC is 70.29%.

As shown in Table 6, the dump truck satisfies the methanol consumption limit of 80 L/100 km under the CHTC-D and C-WTVC cycles. Additionally, according to reference [7], the diesel consumption limit for the dump truck with a gross weight of 25 tons under the CHTC-D cycle is 37.5 L/100 km. In this study, the equivalent

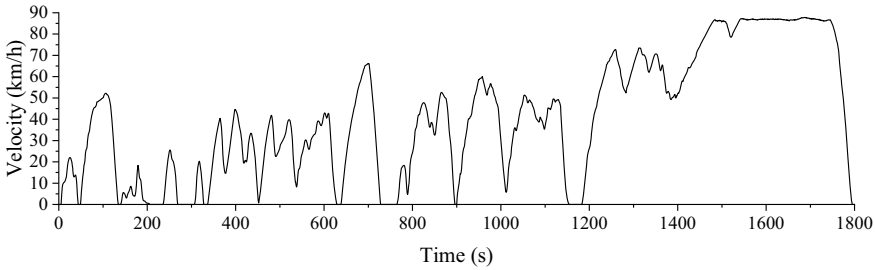


Fig. 11 C-WTVC cycle

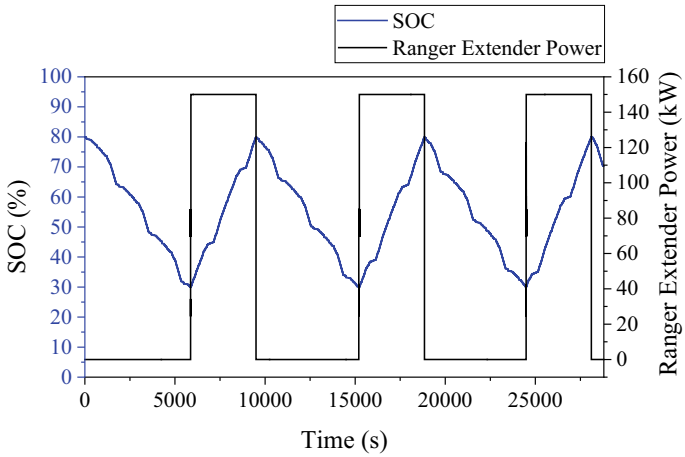


Fig. 12 C-WTVC SOC

Table 6 Simulation results of the fuel economy

Parameters	CHTC-D	C-WTVC
Methanol consumption/(L/100 km)	70.66	77.88
Electric energy consumption/(kWh/km)	1.05	1.17
Equivalent diesel consumption/(L/100 km)	29.79	33.89
Driving range/km	425	385

diesel consumption of the dump truck during the CHTC-D cycle is 29.79 L/100 km, which complies with national standard requirement.

5 Conclusions

(1) A performance test and evaluation of the M100 methanol range extender have been conducted using the bench test. The minimum methanol consumption rate of the range extender is 436.72 g/kWh, while the maximum thermal efficiency is 35.85%.

(2) The design of the parameters for the range extended electric dump truck accords to the basic specifications and performance indicators of the vehicle. By considering the operating environment and the lifespan of the power battery, establish the parameters for the power and drive systems of the range extended electric dump truck.

(3) The operating point of the range extender is determined through performance test. Simultaneously, to enhance fuel economy and satisfy the power demands of the dump truck, a 4-speed AMT with a maximum transmission ratio of 5.0 has been designed, along with its corresponding shift strategy.
(4) Through simulation, the performance of the dump truck satisfies the design requirements. The methanol consumption of the CHTC-D cycle is 70.66 L/100 km, and the equivalent diesel consumption is 79.44% of the national standard limit. Additionally, the price of methanol is approximately 40% that of diesel. Compared to similar diesel-powered dump trucks, the overall cost is lower.
(5) In order to reduce exhaust pollutant emissions from the range extender and comply with the China VI emission standard, the range extender utilizes an equivalent combustion mode and Exhaust Gas Recirculation (EGR), which correspondingly reduces thermal efficiency. Additionally, the range extender operates at a fixed point when generating electricity, all of them impacting the fuel economy of the dump truck. Consequently, the development of control strategies tailored to various driving conditions and power requirements will be the focus of future research.

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